Web-based Nominal Group Technique Decision Making Tool using Blockchain

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Abstract— **The overall interest of modern managers is to find effective group decision-making methods when members with diverse backgrounds and perspectives have to solve problems together. Nominal Group Technique is used as one of the methods for group decision making. However, the Nominal Group Technique is not easy to ensure anonymity in a face-to-face environment, and there is no clear evidence to support the judgment of results after voting. In addition, it is difficult to trust the data because the integrity of the data generated by the voting result is not ensured. In this paper, propose a web-based Nominal Group Technique using blockchain. Web-based Nominal Group Technique can construct an agent - based model and request the blockchain network when data management is needed to guarantee the integrity of the data. In addition, the anonymity of all members is ensured through the web chat function, and the voting results can be visualized using the Priority Graph Tool, which is implemented on the web during the group decision making.**

Keywords— Nominal Group Technique, Blockchain, web-based

I. INTRODUCTION

Group decision making is a type of process in which a problem or situation that arises in a group is analyzed, and alternatives are considered and evaluated to select a solution. Methods that can improve group decision making include Brainstorming, Nominal Group Technique(NGT), and Delphi technique. Among them, the NGT takes into consideration the opinions of everyone in the group and it is a technique to decide the solution through voting. A study using the Nominal Group Technique in group decision making was used to screen and manage pregnant women in the emergency room[1], as a method for group decision making by the Committee[2], it is used to determine the relative efficiency between the two methods using the Interacting Group Technique and the Nominal Group Technique, used to create ideas in the field of Health Services Research (HSR)[3].

However, these papers focus exclusively on the face-to-face environment when using the NGT in group decision making, and it is difficult to provide a clear basis for the support of the voting results. In addition, the integrity of the data according to the voting result is not ensured, making it difficult to trust the data. Therefore, propose an agent-based model that applies blockchain to web environment NGT. Web-based NGT expresses opinions in anonymity-protected computer environment and discusses them through chat. The research of PP Lago shows the results of comparing traditional NGT with web NGT in group decision making. As a result of the comparison, participants felt that the satisfaction of the web environment NGT was similar to the result of the traditional NGT, and the online environment felt less oppressed[4].

In this paper, The web-based NGT communicates with blockchain network with agent - based model and manages data. When the host or the participant inputs the data required for the NGT process, the computer becomes an agent-based model that automatically processes the NGT process according to the data. In addition, it is possible to express opinions in anonymity-protected environment through web chat, and Priority Graph tool, which is based on the DCPT(Distributed Collaboration Priorities Tool) implemented on the web at item voting can be used to express participants voting intentions and voting results clearly through graphs. Therefore, we constructed an agent model that guarantees the integrity of data through blockchain, ensured anonymity through web chat, and visualized the data using the Priority Graph Tool..

This paper is organized as follows. In Section 2, introduce NGT, DCPT, and Blockchain. Section 3 describes the structure and process of web-based NGT. Section 4 presents conclusions and future work.

II. RELATION WORKS

A. Nominal Group Technique

NGT is a technology that quickly generates many ideas from groups that need to make decisions, and after debating, votes are used to prioritize ideas. This technology ensures fair participation of all participants, as well as those who are influential in the group. Therefore, this technique is used to create many ideas, and all participants to minimize the influence of other participants and create opinions freely. It is also used to confirm the priority of the solution or to select some alternatives for further review[5].

NGT consists of the following steps.

- (i) Create an idea for a topic: The facilitator sets the topic, and the participants list their ideas on paper without discussion.
- (ii) Sharing through the idea record: The facilitator records the idea of each group participant on a flip chart so that all participants can see it. Records are kept until

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all ideas are exhausted. Discussion of items is prohibited at this time.

- (iii) Discussion on ideas: Discuss the ideas in the flip chart.
- (iv) Vote on ideas: Each participant votes on ideas based on the knowledge gained from the discussion. The average of this ranking is used as the group's doctor. Here, NGT can make further decisions by ending group decisions, or through repeated discussions and voting.

B. DCPT

DCPT is a system that supports the voting function and the result graph to support the priority of the requirements items used in software collaboration development. Participants in the vote on the graph with x indicating the Importance of each item and y indicating difficulty. As a result of voting, the consensus points of the participants for each item are calculated and displayed on the graph, and a graph showing the priority of the item is output. The priorities are output according to each of the four graph models: Average Model, Ratio Model, Diagonal Model, and Risk Model. These DCPTs are used in the WinWin model and show an example of effective support for group decision making in Park's study[6]. In this paper, I use the DCPT-based Priority Graph Tool when voting on group decision making and substitute DCPT's software requirement items with NGT group items.

C. Blockchain

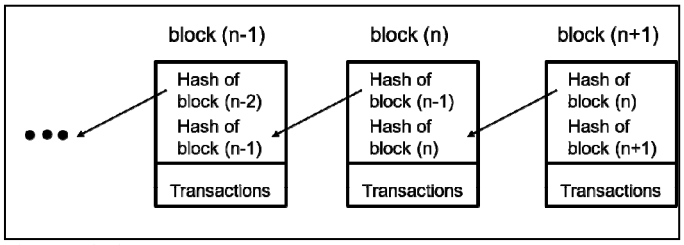

Fig. 1. Block Structure.

The blockchain is a kind of decentralized ledger in which information of transactions occurring in a P2P network environment is shared among P2P network participants to ensure data integrity. Blocks constituting a blockchain are divided into a block header and a block body. The structure is briefly shown in Fig 1. The operation of the blockchain is as follows. When a transaction occurs in a service program, all member nodes determine the validity of the transaction according to a rule determined for each blockchain, that is, a consensus algorithm, generate a block, transmit it to another member node, and update the ledger. In the case of the bit coin, the node that finds the nonce value first through the calculation of the hash value using the PoW consensus algorithm generates the block, and transmits the corresponding block and the nonce value to all the member nodes. The node receiving the block confirms the validity of the block through nonce and constructs a chain by connecting the blocks received in the most recent block of the own ledger. In Fig 1, you can see that the hash value of n-1 block existing in n block is connected to the hash value of previous n-1 block to form a blockchain[7].

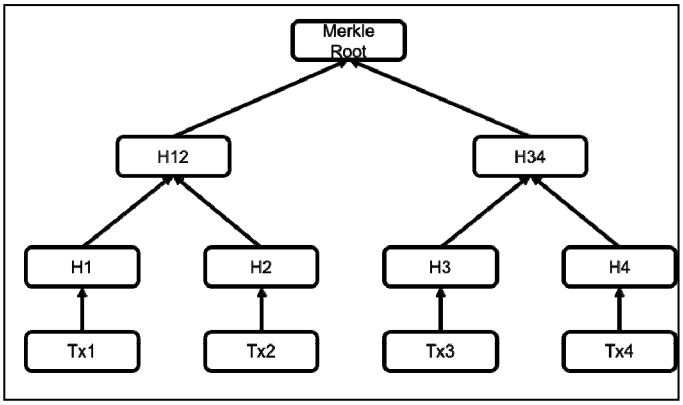

Fig. 2. Merkle Tree.

A merkle tree is used to validate a transaction in a blockchain. The final merkle root value in the merkle tree is included in the block to verify that the transaction is tampered with. The merkle tree is a tree structure consisting of hash values, as shown in Fig 2. The bottom-most node indicates the transaction's data, and the parent nodes are hash values of child nodes. The top node is referred to as the merkle root. Since the merkle route value changes even if only one of the data of the lower node is modulated due to the hash value characteristic of the merkle route, the merkle route can grasp the modulation of the corresponding data[8].

The first example of applying this blockchain technique is a bitcoin[9], an electronic money system that solves the double payment problem caused by the Byzantine general problem[10] in the P2P environment by using the Proof of Work (PoW) agreement algorithm. The web NGT of this paper adopts Hyperledger Fabric[11] which is one of open source blockchain which can be flexibly used considering various environments. Hyperledger Fabric is a Linux Foundation project, a blockchain system for B2B.

III. WEB-BASED NGT

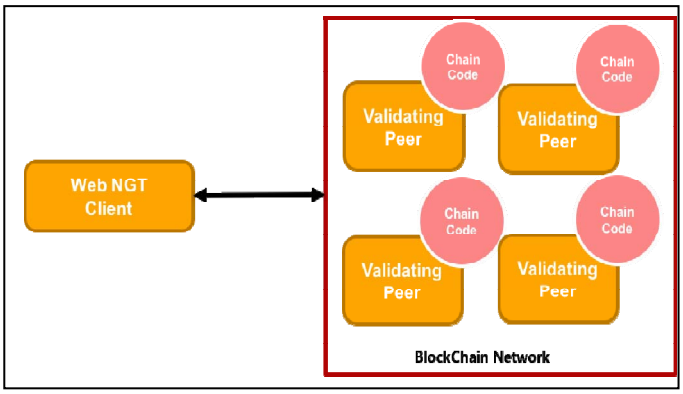

Fig. 3. Web-based NGT Structure.

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A. *Web-based NGT Structure*

The proposed web - based NGT is a group decision tool that can be used in electronic devices such as computers and smart phones capable of the Internet.

The structure of Fig 3 uses Hyperledger Fabric which is one of blockchain open source, and it can be roughly divided into Web NGT Client that provides web service to participants and blockchain Network which stores data for integrity. The web NGT structure that communicates with the blockchain is as follows.

- (i) The Web NGT Client is an application that implements the NGT process. The NGT Client outputs a web page that will be directly viewed by the participants of the group decision making process. It records data in the ledger of the blockchain network . When data is needed, it requests the blockchain network, receives the value, and outputs it to the screen. It is called a transaction request in a blockchain network that a Web NGT Client writes or calls data.
- (ii) In the blockchain Network, there exists a validating peer, which is the same concept as a node normally called in bitcoin. When the Web NGT Client makes a transaction request, the validating peer on the blockchain network checks the validity of the transaction using the agreed algorithm, and creates and records the block in the ledger. The Validating Peer keeps the latest book by always checking each other's ledger. Chaincode is used to create SmartContract in Hyperledger Fabric. When a Web NGT Client requests a transaction, the block chain network operates by business logic implemented in Chaincode. The Chaincode has a function to store a value and retrieve a value when a transaction request is made by the Web NGT Client.

B. *Web-based NGT Process*

The same procedure as the existing NGT process is performed. However, Web-based NGT provides web based anonymity chatting and voting system using Priority Graph Tool, and that designed an agent - based model in which the computer takes over the existing NGT process led by the host. When data is created or changed during the process, data is transmitted to the blockchain network to ensure data integrity.

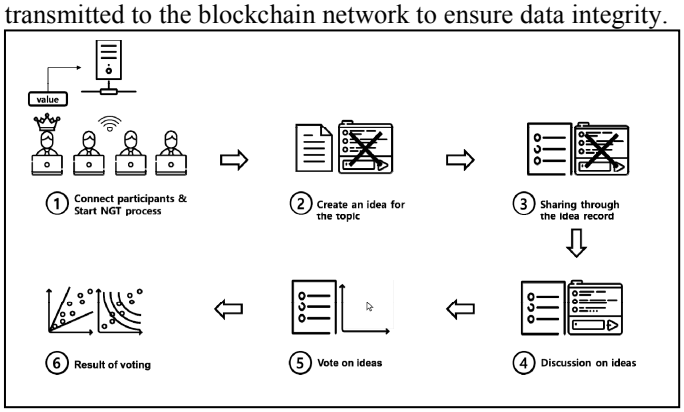

Fig. 4. Web-based NGT Process

In addition, a log of the process progress record can be sent to the blockchain network to track the process history.

Fig.4. shows the process of Web NGT and introduces the process of Web NGT.

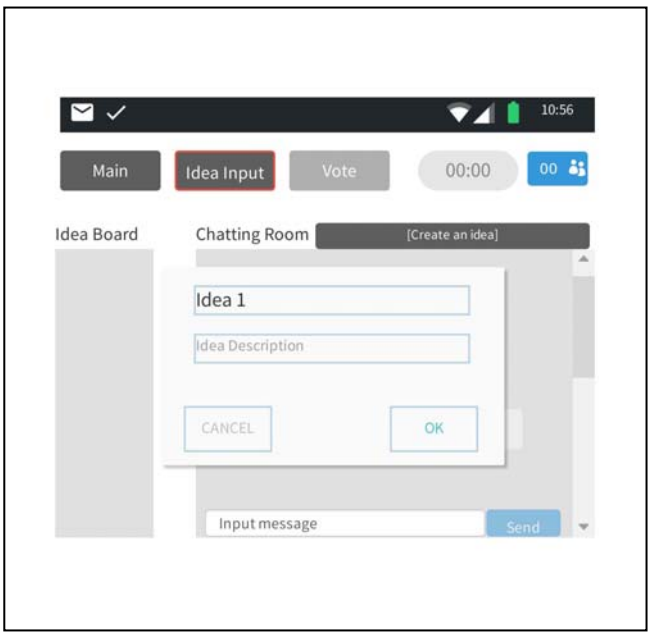

Fig. 5. Create an idea screen

- (i) Connect participants & start NGT process: Participants in the group decision-making process first access the Web NGT. In order for the computer to proceed with the NGT process, the host inputs the topics and precautions, idea creation time, idea discussion time, and voting time as variables, and the computer that receives the variables starts the NGT process. The NGT process records the data in log form and transmits them to the blockchain network. It performs log data transfer at every process step, and distinguishes log data by date and time, NGT process number.
- (ii) Create an idea for the topic: Fig.5 shows the idea create screen on the web. When the NGT starts, the computer displays the topics and precautions of the received variables in the chat window and informs them that it is an idea generation step. During the idea creation time, participants are not allowed to chat, so they can write their own ideas without discussing each other. The idea should be written in the idea input window within one line of each idea title and content.

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- (iii) Sharing through the idea record: After the idea creation time has elapsed, the computer prints the idea of each group participant on an idea board that all participants can view. Give participants 5 minutes to check out their ideas for a smooth discussion. This step also has no chat privileges for participants, just like creating an idea. The idea output on the board is sent to the blockchain network along with the topic.

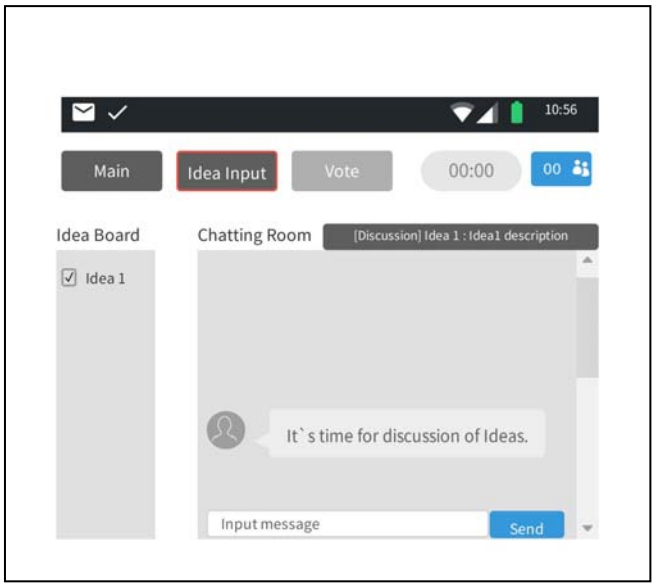

Fig. 6. Discussion on idea screen

- (iv) Discussion on ideas: Fig.6. shows an idea discussion screen on the web. The computer will proceed to discuss ideas in the order in which they appear on the idea board. The title and contents of the ideas to be discussed are displayed, and the participants are given chat privileges during the idea discussion time so that the discussion can proceed. Idea Discussion is conducted during the discussion time designated by the facilitator for one idea, and the discussion is repeated until all ideas are exhausted. Like other NGT process steps, the discussion data is sent to the blockchain network to record the idea discussion process.

- (v) Vote on ideas: Fig.7 shows an idea voting screen on the web. After discussing the idea, the participant uses the Priority Graph Tool to vote on the X-axis, which represents the risk of each idea, and the Y-axis, which represents importance. The values that are voted on the idea make a transaction request and write the data to the ledger of the blockchain network.

- (vi) Result of voting: At the end of the voting time, data is fetched from the ledger of the blockchain network through a transaction request to the blockchain network. The data is used to output a consensus graph and a priority graph according to the voting result of each participant on the web through a calculation process, and the graph visualizes the decision of the group.

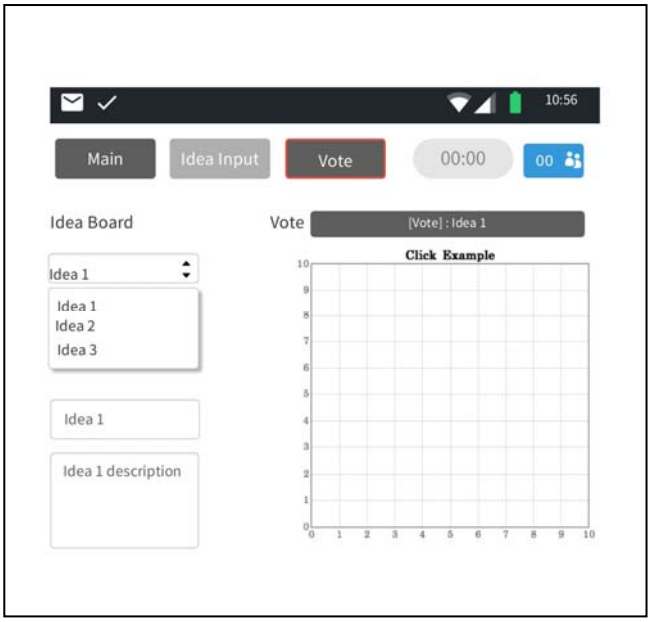

Fig. 7. Vote on idea screen

IV. CONCLUSION AND FUTURE WORKS

When NGT is used in group decision making. First, it is not easy to guarantee anonymity in face to face environment. Secondly, it is difficult for the participants to have a clear basis to support the judgment of the result after voting. Thirdly, since the data generated by the voting result does not guarantee the integrity of the data, it is difficult to trust the data. Therefore, we proposed web environment NGT using blockchain. The proposed NGT is based on the agent model, and the process is automatically performed in an environment that guarantees the anonymity. Using the ledger of the blockchain network, the integrity of the data according to the voting result is ensured and the voting result is visualized through the graph. However, since only the normal situation of the NGT process is considered, if the problem of the proposed model process occurs, the computer can`t actively cope with it. In addition, when using a blockchain network, only the reading and writing functions of the voting result data are used in a limited manner. Therefore, future work will study various situations of web NGT to be able to prepare for abnormal situation by testing and considering. In addition, we will make it possible to calculate the average of the voting data in the blockchain network and extend the blockchain network to be viewed by the web-based NGT so that the participants can verify the integrity of the data.

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